

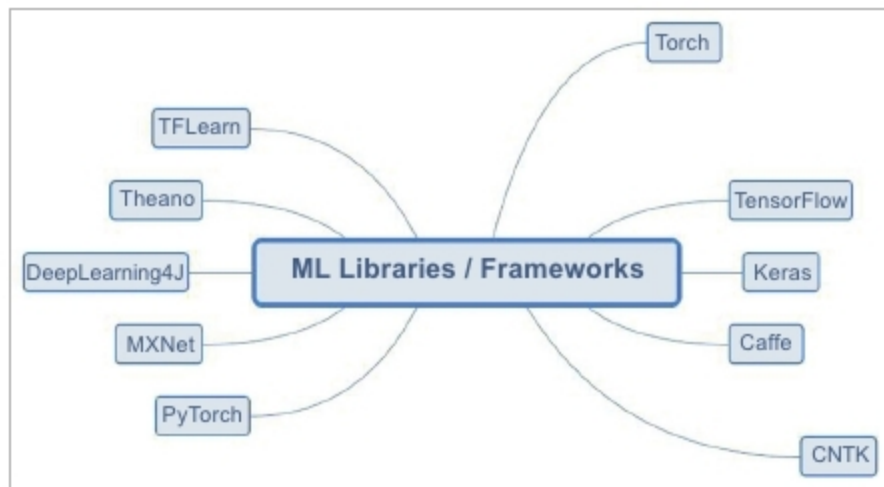


Figure 1: Machine learning frameworks/libraries

Installation

The pre-requisite for the installation of Apache SystemML is Apache Spark. The variable `SPARK_HOME` should be set to the location where Spark is installed.

Installing Apache SystemML for the Python environment can be done with the Pip command as shown below:

```
pip install systemml
```

More information about this can be accessed at <http://systemml.apache.org/docs/1.2.0/index.html>.

If you want to work with the Jupyter Notebook, the configuration can be done as follows:

```
PYSPARK_DRIVER_PYTHON=jupyter PYSPARK_DRIVER_PYTHON_
OPTS="notebook" pyspark --master local[*] --conf "spark.
driver.memory=12g" --conf spark.driver.maxResultSize=0 --conf
spark.default.parallelism=100
```

Instructions for installing SystemML with Scala can be got from the official documentation at <http://systemml.apache.org/install-systemml.html>.

DML and PyDML

As stated earlier, flexibility is another advantage of SystemML, which it achieves through a high level declarative machine learning language. This language ships in two flavours—one is called DML and has syntax like R. The other is PyDML, which is like Python.

A code snippet of PyDML is shown below:

```
aFloat = 3.0
bInt = 2
print('aFloat = ' + aFloat)
print('bInt = ' + bInt)
print('aFloat + bInt = ' + (aFloat + bInt))
print('bInt ** 3 = ' + (bInt ** 3))
print('aFloat ** 2 = ' + (aFloat ** 2))

cBool = True
print('cBool = ' + cBool)
```

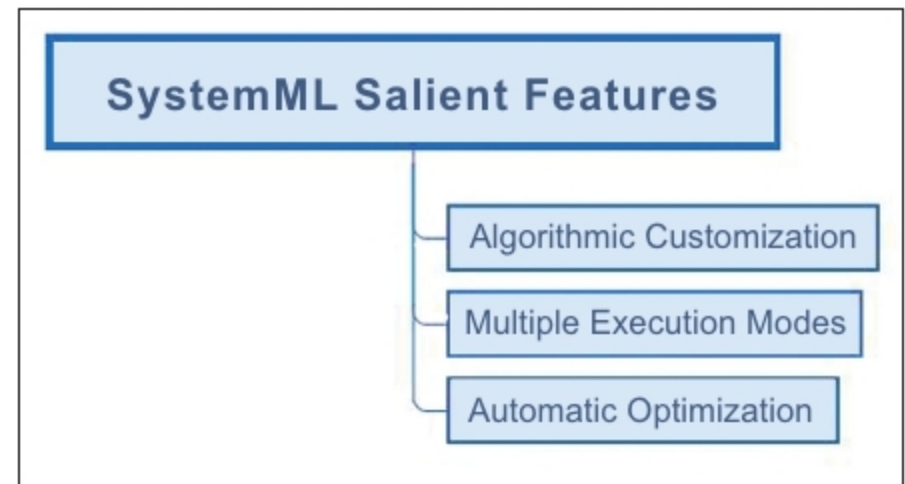


Figure 2: SystemML's salient features

```
print('(2 < 1) = ' + (2 < 1))

dStr = 'Open Source'
eStr = dStr + ' For You'
print('dStr = ' + dStr)
print('eStr = ' + eStr)
```

A sample code snippet of DML is shown below:

```
aDouble = 3.0
bInteger = 2
print('aDouble = ' + aDouble)
print('bInteger = ' + bInteger)
print('aDouble + bInteger = ' + (aDouble + bInteger))
print('bInteger ^ 3 = ' + (bInteger ^ 3))
print('aDouble ^ 2 = ' + (aDouble ^ 2))

cBoolean = TRUE
print('cBoolean = ' + cBoolean)
print('(2 < 1) = ' + (2 < 1))

dString = 'Open Source'
eString = dString + ' For You'
print('dString = ' + dString)
print('eString = ' + eString)
```

A basic matrix operation with PyDML is shown below:

```
A = full("1 2 3 4 5 6", rows=3, cols=2)
print(toString(A))

B = A + 4
B = transpose(B)
print(toString(B))

C = dot(A, B)
print(toString(C))

D = full(5, rows=nrow(C), cols=ncol(C))
D = (C - D) / 2
print(toString(D))
```